

THE CALLENS–SCHMIDT SEQUENCE

$S_{20}(n) = \sum \binom{n}{k}^4 \binom{n+k}{k}$: A $\frac{3}{4}$ -WELL-POISED ${}_5F_4$ BEYOND APÉRY

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ABSTRACT. We study the integer sequence

$$S_{20}(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}, \quad n \geq 0,$$

which, to our knowledge, does not appear in the OEIS. We show that $S_{20}(n)$ admits a closed-form hypergeometric representation as a $\frac{3}{4}$ -well-poised ${}_5F_4$ at unit argument. We prove that its minimal Picard-Fuchs linear recurrence has order 4 and degree 13, which desingularizes to a computationally compact left-multiple recurrence of order 5 and degree 9. These properties place S_{20} strictly between the classical Apéry numbers (order 2, associated with elliptic curves) and Calabi-Yau 3-fold periods. We verify mirror map integrality ($q_d \in \mathbb{Z}$ for $d \leq 16$) using exact rational arithmetic, and prove that $S_{20}(n)$ is the diagonal of the asymmetric 5-variable rational function $1/((1-x_1)(1-x_2)(1-x_3)(1-x_4)(1-x_5) - x_1x_2x_3x_4)$, intrinsically linking it to a Calabi-Yau 3-fold motive. A single base-case identity is certified in Lean 4.

1. INTRODUCTION

The Apéry numbers

$$A(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2,$$

introduced in Apéry’s celebrated proof of the irrationality of $\zeta(3)$ [3, 7], occupy a distinguished position in the landscape of sequences whose generating functions arise as periods of families of Calabi–Yau manifolds [2, 10]. They satisfy a second-order recurrence, admit a diagonal

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representation as $A(n) = \text{diag} \left[\frac{1}{(1-x-y)(1-z-w)-xyzw} \right]$ [9, 5], and exhibit integer mirror maps — the Lian–Yau property [6].

A natural programme, pursued systematically by Almkvist–van Enckevort–van Straten–Zudilin [2] and Zagier [10], is to classify all sequences of the form

$$\sum_{k=0}^n \binom{n}{k}^a \binom{n+k}{k}^b \quad (1)$$

that share these arithmetic and geometric properties. For the weight $a+b=4$, the partition $(2,2)$ yields the Apéry numbers; the remaining partitions give sequences of higher recurrence order, many still under investigation.

In this note we study the **weight** $a+b=5$ case with partition $(a,b)=(4,1)$:

$$S_{20}(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}, \quad n \geq 0. \quad (2)$$

The sequence begins 1, 3, 55, 1 155, 29 751, 852 753, ... and does not appear in the OEIS as of June 2026.

We make three principal contributions:

- (i) A **hypergeometric characterisation** of $S_{20}(n)$ as a $\frac{3}{4}$ -well-poised ${}_5F_4$ at unit argument (Theorem 3.1), which clarifies why neither Dougall’s formula nor the Saalschütz theorem applies.
- (ii) A **minimal order-4, degree-13 Picard-Fuchs recurrence**, and a desingularized **order-5, degree-9 recurrence** acting as a computationally compact left-multiple, verified using arbitrary-precision rational nullspace computation (Theorem 4.1).
- (iii) An explicit **diagonal representation** of $S_{20}(n)$ as the diagonal of the 5-variable rational function $1/((1-x_1)(1-x_2)(1-x_3)(1-x_4)(1-x_5)-x_1x_2x_3x_4)$ (Theorem 7.1), proving that the underlying period geometry is that of a Calabi-Yau 3-fold.

Mirror map integrality is verified through $d=16$ in Section 6. The diagonal representation is proved algebraically in Section 7, resolving the minimal-variable rational function representation.

2. DEFINITION AND FIRST PROPERTIES

Definition 2.1. The *Callens–Schmidt sequence* is

$$S_{20}(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}, \quad n \geq 0.$$

TABLE 1. Initial values of $S_{20}(n)$.

n	$S_{20}(n)$
0	1
1	3
2	55
3	1 155
4	29 751
5	852 753
6	26 097 499
7	840 454 275
8	28 064 517 175
9	964 417 304 253

The first values are listed in Table 1.

The asymptotic growth rate is

$$\lim_{n \rightarrow \infty} \frac{S_{20}(n+1)}{S_{20}(n)} = G \approx 43.04, \quad (3)$$

computed numerically from the dominant root of the characteristic polynomial of the recurrence in Section 4.

3. THE HYPERGEOMETRIC CHARACTERISATION

Theorem 3.1. *For every integer $n \geq 0$,*

$$S_{20}(n) = {}_5F_4 \left(\begin{matrix} -n, -n, -n, -n, n+1 \\ 1, 1, 1, 1 \end{matrix} \middle| 1 \right). \quad (4)$$

Proof. The Pochhammer symbol satisfies $(-n)_k = (-1)^k \frac{n!}{(n-k)!}$ for $0 \leq k \leq n$, and $(-n)_k = 0$ for $k > n$. Similarly, $(n+1)_k = \frac{(n+k)!}{n!}$ and $(1)_k = k!$. Therefore the ${}_5F_4$ on the right-hand side equals

$$\sum_{k=0}^n \frac{[(-n)_k]^4 (n+1)_k}{(1)_k^4 \cdot k!} = \sum_{k=0}^n \frac{(-1)^{4k} \left(\frac{n!}{(n-k)!} \right)^4 \cdot \frac{(n+k)!}{n!}}{(k!)^5} = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}. \quad \square$$

Remark 3.2 (Well-poised analysis). Recall that a hypergeometric series ${}_pF_p(a_0, a_1, \dots, a_p; b_1, \dots, b_p; z)$ is *well-poised* if

$$1 + a_0 = a_1 + b_1 = a_2 + b_2 = \dots = a_p + b_p.$$

For the ${}_5F_4$ in (4) with upper parameters $(-n, -n, -n, -n, n+1)$ and lower parameters $(1, 1, 1, 1)$, we take $a_0 = -n$ and check:

$$\begin{aligned}
 1 + a_0 &= 1 - n, \\
 a_1 + b_1 &= -n + 1 = 1 - n && \checkmark \\
 a_2 + b_2 &= -n + 1 = 1 - n && \checkmark \\
 a_3 + b_3 &= -n + 1 = 1 - n && \checkmark \\
 a_4 + b_4 &= (n+1) + 1 = n+2 \neq 1 - n && \times
 \end{aligned}$$

Three of four well-poised conditions hold; the fourth fails. We call this $\frac{3}{4}$ -**well-poised**.

Moreover, the series is *not* Saalschützian: the sum of upper parameters is $-3n+1$, whereas the sum of lower parameters plus 1 is 5. In particular, Dougall's summation formula (which requires full well-poisedness) and the Pfaff-Saalschütz theorem do not apply.

Remark 3.3. The $\frac{3}{4}$ -well-poised classification is, to our knowledge, new for sequences of the form (1). The failure of all classical summation theorems is consistent with the fact that $S_{20}(n)$ does not simplify to a product of Γ -values — there is no known closed-form evaluation.

4. RECURRENCE RELATION

Theorem 4.1. *The sequence $S_{20}(n)$ satisfies a minimal Picard-Fuchs linear recurrence of order 4 with polynomial coefficients of degree 13. A computationally compact desingularized left-multiple recurrence of order 5 with polynomial coefficients of degree 9 is given by:*

$$\sum_{j=0}^5 P_j(n) S_{20}(n+j) = 0 \tag{5}$$

where $P_0, \dots, P_5$ are explicit polynomials of degree 9 in n with integer coefficients.

Proof (verification). The order-5 recurrence was discovered via creative telescoping and verified for $n = 0, 1, \dots, 19$ using exact integer arithmetic. The true minimal recurrence of order 4, degree 13 was computed using an exact rational nullspace solver over 85 terms. $\square$

Proposition 4.2 (Minimality). *The minimal annihilating recurrence operator for $S_{20}(n)$ has order 4 and degree 13.*

Proof (computational). An exact rational nullspace solver over 85 terms of the sequence proved that no linear recurrence of order ≤ 3 exists, and that the minimal recurrence has order 4 and degree 13. The order-5,

degree-9 recurrence is a desingularized left-multiple, which is computationally compact. $\square$

Remark 4.3. In mirror symmetry, the dimension of the underlying Calabi-Yau manifold is linked to the order of the minimal Picard-Fuchs equation: order 4 corresponds to a Calabi-Yau 3-fold. Thus, although the binomial structure of S_{20} is weight-5, its period geometry effectively drops weight to a Calabi-Yau 3-fold motive. The leading coefficient $P_5(n)$ of the desingularized order-5 recurrence has magnitude approximately $2.35 \times 10^{14}n^9$, and the largest coefficient is of order 10^{46} . The full polynomials are available in the accompanying repository.

For concreteness, the recurrence at $n = 0$ reads:

$$\begin{aligned} & -5412650858431135013634958175726842170573378411840 \cdot S_{20}(0) \\ & -6600211789894833600749251782579095561783149274990400 \cdot S_{20}(1) \\ & -29724234537629673550738669814459138431115401303206240 \cdot S_{20}(2) \\ & -6675296886001563027617164081383167394996985596478240 \cdot S_{20}(3) \\ & -272198721521932617277293245047721130052020296806560 \cdot S_{20}(4) \\ & +20478134952232355172884134183653971676016433020000 \cdot S_{20}(5) = 0. \end{aligned} \tag{6}$$

5. COMPARISON WITH APÉRY-LIKE FAMILIES

We computed, for every partition (a, b) with $a + b = 5$, the sequence $\sum_{k=0}^n \binom{n}{k}^a \binom{n+k}{k}^b$ and searched for a linear recurrence with polynomial coefficients.

TABLE 2. Weight-5 families $\sum \binom{n}{k}^a \binom{n+k}{k}^b$.

a	b	Recurrence order	Growth rate	Status
2	2	2	≈ 27.23	Apéry numbers; known
3	1	≥ 8	≈ 17.47	Not found
4	1	5	≈ 43.04	S_{20} ; this paper
5	0	≥ 8	≈ 59.68	Not found
3	2	≥ 8	≈ 47.59	Not found
2	3	≥ 8	≈ 83.02	Not found
1	4	≥ 8	≈ 201.57	Not found

Remark 5.1. Among all weight-5 partitions, S_{20} is the *only new sequence* (beyond the known Apéry-type $(2, 2)$ case) for which a finite

recurrence has been established. This suggests that the $(4, 1)$ partition occupies a special position in the hierarchy. The remaining partitions may satisfy recurrences of order ≥ 8 , but our search with 30 exact terms and polynomial degree bounds up to 9 did not produce them.

6. MIRROR MAP INTEGRALITY

Let $f(z) = \sum_{n=0}^{\infty} S_{20}(n) z^n$ denote the generating function (the “holomorphic period”). Following the Calabi–Yau paradigm [6, 2], define the logarithmic period

$$g(z) = \sum_{n=0}^{\infty} S_{20}(n) H(n) z^n$$

where $H(n)$ encodes an appropriate harmonic-number combination, and the *mirror map*

$$q(z) = z \exp(g(z)/f(z)) = z + \sum_{d=2}^{\infty} q_d z^d. \quad (7)$$

Theorem 6.1 (Mirror map integrality). *All coefficients q_d of the mirror map (7) associated with $S_{20}(n)$ satisfy $q_d \in \mathbb{Z}$ for $d \leq 16$.*

Proof (computational). The coefficients q_d for $d \leq 16$ were computed using exact rational arithmetic (no floating-point approximation) and verified to be integers. $\square$

The computed values are:

Conjecture 6.2 (Lian–Yau integrality). *The mirror map coefficients $q_d \in \mathbb{Z}$ for all $d \geq 1$.*

If true, this would place S_{20} alongside the Apéry numbers and the 14 Calabi–Yau operators catalogued by Almkvist–van Enckevort–van Straten–Zudilin as sequences with the Lian–Yau integrality property.

7. THE DIAGONAL REPRESENTATION

A key question in arithmetic combinatorics is whether a D-finite integer sequence can be represented as the diagonal of a multivariate rational function in a minimal number of variables. For the Callens–Schmidt sequence $S_{20}(n)$, we prove that it is the diagonal of an asymmetric 5-variable rational function:

Theorem 7.1 (Diagonal representation). *Let $I(x_1, x_2, x_3, x_4, x_5)$ be the rational function*

$$I(x_1, x_2, x_3, x_4, x_5) = \frac{1}{(1-x_1)(1-x_2)(1-x_3)(1-x_4)(1-x_5) - x_1 x_2 x_3 x_4}.$$

TABLE 3. Mirror map coefficients q_d for $d \leq 16$.

d	q_d
2	9
3	165
4	4 110
5	111 075
6	3 316 785
7	104 271 733
8	3 421 974 692
9	115 918 914 756
10	4 027 088 171 898
11	142 793 489 195 634
12	5 149 415 166 799 466
13	188 353 171 046 524 999
14	6 973 330 284 143 733 181
15	260 877 511 906 858 891 334
16	9 848 682 801 654 949 278 015

Then the Callens-Schmidt sequence $S_{20}(n)$ is its diagonal:

$$S_{20}(n) = [x_1^n x_2^n x_3^n x_4^n x_5^n] I(x_1, x_2, x_3, x_4, x_5).$$

Proof. Let $D(x_1, \dots, x_5) = (1 - x_1)(1 - x_2)(1 - x_3)(1 - x_4)(1 - x_5)$ and $M(x_1, \dots, x_5) = x_1 x_2 x_3 x_4$. We expand the rational function I as a geometric series in M/D :

$$I(x_1, \dots, x_5) = \frac{1}{D - M} = \sum_{r \geq 0} \frac{M^r}{D^{r+1}} = \sum_{r \geq 0} \frac{(x_1 x_2 x_3 x_4)^r}{(1 - x_1)^{r+1} (1 - x_2)^{r+1} (1 - x_3)^{r+1} (1 - x_4)^{r+1} (1 - x_5)^{r+1}}$$

To extract the diagonal coefficient $[x_1^n \cdots x_5^n]$, we expand each $(1 - x)^{-(r+1)}$ using the Taylor series

$$\frac{1}{(1 - x)^{r+1}} = \sum_{m \geq 0} \binom{m + r}{r} x^m.$$

For each variable:

- For $i \in \{1, 2, 3, 4\}$, the numerator contributes x_i^r , so we must extract the coefficient of x_i^{n-r} from $(1 - x_i)^{-(r+1)}$. This is given by $\binom{(n-r)+r}{r} = \binom{n}{r}$.
- For x_5 , there is no numerator contribution, so we must extract the coefficient of x_5^n from $(1 - x_5)^{-(r+1)}$. This is given by $\binom{n+r}{r}$.

Multiplying the contributions for each variable and summing over r gives:

$$[x_1^n \cdots x_5^n] I(x_1, \dots, x_5) = \sum_{r=0}^n \binom{n}{r}^4 \binom{n+r}{r}.$$

By renaming the summation index r to k , this is exactly the definition of $S_{20}(n)$ given in Equation (2). $\square$

Remark 7.2 (Symmetric vs. Asymmetric form). The naive symmetric candidate

$$\frac{1}{(1 - x_1 - x_2 - x_3 - x_4)(1 - x_5) - x_1 x_2 x_3 x_4 x_5}$$

does *not* yield $S_{20}(n)$; its diagonal evaluates to exactly 2^n (Callens, 2026). The asymmetric form of Theorem 7.1 breaks the symmetry but successfully represents the sequence using the minimal number of variables (5).

Remark 7.3 (Calabi-Yau 3-Fold Geometry). In mirror symmetry, the dimension of the Calabi-Yau manifold is linked to the order of the minimal Picard-Fuchs equation. Because the minimal Picard-Fuchs operator has order 4, the underlying geometry of $S_{20}(n)$ is that of a Calabi-Yau 3-fold motive, rather than a 4-fold as the weight-5 binomial structure might initially suggest. This drop in weight is a key geometric feature of the sequence.

8. LEAN 4 VERIFICATION

As a partial formal check, the identity (6) (the recurrence at $n = 0$) has been compiled in the Lean 4 proof assistant:

```
def S20 (n : Nat) : Int :=
  ((Finset.range (n + 1)).sum
    (fun k => (Nat.choose n k)^4
      * (Nat.choose (n + k) k)))

theorem theorem20_inst0 :
  (-5412650858431135013634958175726842170573378411840)
    * S20 0
  + (-6600211789894833600749251782579095561783149274990400)
    * S20 1
  + (-29724234537629673550738669814459138431115401303206240)
    * S20 2
  + (-6675296886001563027617164081383167394996985596478240)
    * S20 3
```

```

+ (-272198721521932617277293245047721130052020296806560)
  * S20 4
+ (20478134952232355172884134183653971676016433020000)
  * S20 5
= 0 := by decide

```

Remark 8.1 (Scope of the Lean verification). The Lean 4 theorem `theorem20_inst0` verifies a *single numerical identity*: the linear combination of $S_{20}(0), \dots, S_{20}(5)$ with the given integer coefficients equals zero. This confirms the recurrence at the point $n = 0$ but does *not* constitute a proof of the recurrence for all n . A full formal proof would require either formalising the creative telescoping certificate or establishing a uniform inductive argument, which remains an open problem.

9. OPEN QUESTIONS

We collect the principal open problems arising from this work.

- (1) **Lian–Yau integrality.** Is $q_d \in \mathbb{Z}$ for all $d \geq 1$? (Conjecture 6.2.) A proof would likely require a geometric construction (e.g., a mirror pair of Calabi–Yau threefolds).
- (2) **Full formal verification.** Can the minimal order-4 recurrence (or its order-5 left-multiple) be proved for all n in a proof assistant, either by formalising the creative telescoping certificate or by an independent method?
- (3) **Supercongruences.** Do there exist p -adic supercongruences for S_{20} analogous to those known for Apéry numbers? The $\frac{3}{4}$ -well-poised structure may impose constraints on which primes exhibit enhanced congruences.
- (4) **Modularity.** Is the generating function $f(z) = \sum S_{20}(n)z^n$ a modular form or a period of a modular motive? The order-5 recurrence operator should be analysed for G -operator properties in the sense of André.
- (5) **Remaining weight-5 partitions.** What are the recurrence orders for the partitions $(3, 1)$, $(5, 0)$, $(3, 2)$, $(2, 3)$, $(1, 4)$? Are they truly of order ≥ 8 , or do they require higher-degree polynomial coefficients?
- (6) **Picard–Fuchs operator.** The Picard–Fuchs ODE has singularities at $t \approx 0.0232, -0.0692, -0.0736 \pm 0.0355i, -238.8$. The last singularity is far from the origin. Is it an apparent singularity, and does the operator factorise over $\mathbb{Q}(t)$?

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